

IT549: Deep Learning – Lab 3 Assignment

Image-Based AQI Classification using CNN and Pretrained Models

Dataset: <https://drive.google.com/drive/folders/1u-sBxgNB67GfhCQ2f7xRkDIF6fglZZrP?usp=sharing>

Submission

Public GitHub repository link only.

Objective

Demonstrate the use of deep learning for image classification by predicting AQI_Class from images of locations.

You will implement and compare:

1. A basic CNN model trained from scratch
2. A pretrained convolutional neural network (transfer learning)

The assignment focuses on building a complete image classification pipeline including preprocessing, model training, evaluation, and result interpretation.

Dataset Scope

Dataset consists of 2 parts:

- data.csv – file with image path and corresponding features
- sampled_images – images from various classes

You must use only the following fields from data.csv:

- image_path (input image)
- AQI_Class (target class)

Task 1 – Data Preparation

- Load the dataset containing image paths and AQI class labels.
- Resize images to a consistent resolution (e.g., 224×224).
- Normalize pixel values.
- Create reproducible train/validation/test splits (suggestion: 70 / 15 / 15).

Task 2 – Basic CNN Model

- Implement a Convolutional Neural Network from scratch.
- Train the model to classify images into AQI_Class categories.
- Report classification performance on the test set.

Task 3 – Pretrained CNN Model (Transfer Learning)

- Use a pretrained model such as ResNet, VGG, or EfficientNet.
- Replace the final classification layer to match the number of AQI classes.
- Train the model using transfer learning.
- Compare performance with the basic CNN model.

Task 4 – Model Training and Evaluation

Evaluate models using:

Accuracy, Precision, Recall, F1-score

Generate a confusion matrix for the test dataset and compare performance between the CNN model and the pretrained model.

Task 5 – Training Curves

Plot the following graphs:

- Epoch vs Training Loss, Validation Loss, Training Accuracy, Validation Accuracy

Discuss why pretrained models often outperform models trained from scratch and whether transfer learning helped in this dataset.

Task 6 – Misclassification Analysis

Identify 5–10 misclassified images from the test set.

Visualize them along with predicted class and actual class.

Provide a short explanation for possible reasons behind misclassification.

Deliverables (GitHub Only)

Your public GitHub repository must include:

- A README containing assignment title, your name, and your ID along with the project details.
- Complete runnable code covering preprocessing, embeddings, training, evaluation, and analysis.
- A results summary comparing performance across different input columns.